AI-Driven Loan Default Prediction Using Banking Analytics

Explainable Machine Learning for Credit Risk and Default Prediction in Banking

1. Refined Title

AI-Driven Loan Default Prediction Using Banking Analytics: Explainable Machine Learning for Credit Risk Assessment

The title reflects the two core threads carried through the project: banking-analytics-driven loan default prediction, and explainability as a first-class requirement rather than an afterthought — a distinction the literature survey below shows is still uncommon.

2. Problem Statement

Manual underwriting and legacy scorecards cannot capture the complex, non-linear relationships between borrower behaviour, repayment history, and macroeconomic conditions. Traditional models routinely misclassify risk, directly inflating non-performing assets (NPAs) and reducing capital efficiency for banks and NBFCs. Where institutions do adopt machine learning, most models offer no explanation for a decision, making them hard to trust, audit, or defend to a regulator. Default cases are also a small minority of records, and most studies validate on a single dataset, limiting how well findings generalise across lending contexts.

3. Literature Survey

Eleven recent studies (2024–2026) from IEEE, ACM, and Springer venues were reviewed:

#	Study & Venue	Focus / Contribution
1	Aruleba & Sun (2024), IEEE Access, vol. 12	Ensemble classifiers (Random Forest, AdaBoost, XGBoost, LightGBM) combined with SMOTE-ENN resampling and SHAP explanations on German and Australian credit datasets.
2	Zhu et al. (2024), IEEE ICETCI	LightGBM–XGBoost–local-ensemble stacking for credit default prediction; ensembling lifts detection over single learners.
3	Vihurskyi (2024), IEEE ICDCECE	Explainable AI applied to credit-card fraud/default detection to improve interpretability and analyst trust in flagged cases.
4	Gonaygunta et al. (2025), IEEE SIEDS, pp. 444–449	Interpretable credit-scoring models using explainable AI to empower end users in financial risk assessment.
5	Hartomo, Arthur & Nataliani (2025), IEEE Access	Weighted-loss TabTransformer integrating explainability for imbalanced credit-risk datasets.
6	ACM GAIIS (2025)	Seven-model comparison (Decision Tree, Random Forest, AdaBoost, Bagging, XGBoost, LightGBM, CatBoost); CatBoost reported best on

#	Study & Venue	Focus / Contribution
		AUC.
7	ACM CSAIDE (2025)	Explainable ML on the Home Credit dataset (1.53M records, 3.14% default rate) for consumer-finance risk.
8	Demma Wube et al. (2024), Springer CCIS vol. 2069	Systematic review of deep-learning vs. classical ML techniques for credit scoring, PanAfriConAI 2023.
9	Computational Economics (2025), Springer, DOI 10.1007/s10614-025-10962-9	XGBoost with post-hoc SHAP explanations for early loan-default prediction on P2P lending data.
10	Liang et al. (2025), Computational Economics, Springer	Explainable deep learning for credit-risk analysis of technology-based small and micro enterprises.
11	Kofidis, Cocianu & Uscatu (2026), Springer LNNS	Comparative study of balancing techniques (SMOTE family) across CNN, LSTM, Random Forest, and SVM for loan default.

Consistent pattern across all 11 studies: strong single-model or single-dataset accuracy gains, but explainability is usually bolted on to one "winning" model rather than compared across models, datasets, or lending contexts.

4. Research Gaps Identified

- Single-dataset validation – Papers 1, 3, 5, 6, 9, 10 each validate on one dataset (German credit, Home Credit, or a single P2P platform) — findings rarely transfer across lending environments.
- Explainability applied late – SHAP/LIME (papers 1, 3, 5, 6, 7, 9) is typically bolted on to a single "winning" model after selection, not used to compare candidate models during development.
- Accuracy–interpretability trade-off – Deep/transformer models (papers 5, 10) push accuracy up, but interpretability studies (paper 8) show this often comes at the cost of transparency for non-technical stakeholders.
- No decision-support layer – Almost none of the surveyed work (papers 1–11) packages results into a dashboard loan officers can actually use — most stop at reporting Accuracy/F1/AUC tables.

5. Objectives

- Consolidate benchmark data – Integrate Lending Club, Statlog German Credit, and Home Credit datasets into one structured, analysis-ready pipeline using schema alignment and unified feature encoding.
- Perform deep, decision-grade EDA – Run 12+ analytical techniques — missingness, outliers, class-imbalance, and correlation analysis — before any modelling, using univariate and multivariate statistical methods.

- Engineer risk-aware features – Derive debt-to-income ratio, credit-utilization, and repayment-consistency features using domain-driven feature engineering to sharpen predictive signal.
- Benchmark five ML models – Train and compare Logistic Regression, Decision Tree, Random Forest, XGBoost, and CatBoost under identical, cross-validated conditions.
- Make predictions explainable – Apply SHAP and LIME to the best model so every default-risk score has an auditable, human-readable reason.
- Deliver a usable dashboard – Package results into a visualization dashboard so loan officers see the prediction and the reasoning together.

6. Novelty / Innovation, Feasibility, and Project Plan

6.1 Novelty / Innovation

Each identified gap maps to a specific design decision that is new relative to the surveyed work:

- Single-dataset validation → Three benchmark datasets combined — Lending Club, Statlog German Credit, Home Credit — spanning traditional, classical, and thin-file/alternative lending.
- Explainability applied late → SHAP and LIME run across all five candidate models during comparison, not only on the final chosen model.
- Accuracy vs. interpretability trade-off → Five models spanning simple-to-complex (Logistic Regression → CatBoost) benchmarked on the same metrics, letting stakeholders choose their own point on the trade-off curve.
- No decision-support layer → A visualization dashboard (heatmaps, boxplots, feature-importance charts) turns model output into something a loan officer can act on, not just a metrics table.

6.2 Feasibility

- Data feasibility – 3 datasets are public, well-documented, and widely cited in credit-risk research; combined size supports robust train/test/validation splits.
- Technical feasibility – All five ML models have mature, well-supported Python libraries; SHAP and LIME are standard, actively maintained explainability toolkits; class imbalance is addressable (SMOTE, class-weighting).
- Economic feasibility – Built entirely on open-source tools and free/public datasets — no licensing cost.
- Time feasibility – Scoped to fit the semester timeline with buffer before final submission; a modular pipeline lets EDA, modelling, and XAI stages proceed in parallel.

6.3 Project Plan

Phase	Milestone	Key Activities
1	Data Acquisition & Cleaning	Collect and merge 3 datasets, handle missing values, duplicates, encoding.
2	Exploratory Data Analysis	12+ techniques: univariate through multivariate, correlation,

Phase	Milestone	Key Activities
		imbalance analysis.
3	Feature Engineering & Selection	DTI ratio, credit utilization, mutual information, RFE.
4	Model Building & Comparison	Train LR, Decision Tree, Random Forest, XGBoost, CatBoost.
5	Explainability & Evaluation	SHAP + LIME analysis; Accuracy/Precision/Recall/F1/ROC-AUC.
6	Dashboard & Final Reporting	Build visualization dashboard; consolidate findings for submission.

7. Team Details

Role	Name
Team Member	Tejasree Sagiraju
Team Member	Sruthi Bhatraju
Team Member	B. Bhavya Sai Sree
Guide	K. Swapnika
Course	Data Analytics & Visualisation – 1 (24DSB3101), A.Y. 2026–2027
Review	Review-1 Presentation · 5 August 2026

References